

HR WORKFORCE ANALYTICS

SQL Business Case Study

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1. EXECUTIVE SUMMARY

For this project, I analysed an HR dataset of 311 employees across 6 departments to answer real workforce questions using SQL. I used Python to load the data into SQLite, then wrote 10 queries — including window functions and CASE statements — to explore salary patterns, attrition, performance, recruitment effectiveness, and absenteeism. The goal was to find insights that an HR team or business manager could actually act on.

Total Employees	Overall Attrition	Avg Salary	Top Dept Attrition
311	33.4%	\$68,940	39.7%
<i>Across 6 departments</i>	<i>104 terminated</i>	<i>Excl. Executive Office</i>	<i>Production dept</i>

2. HEADCOUNT & SALARY BY DEPARTMENT

Business Question: How are employees and salaries distributed across departments?

Department	Headcount	Avg Salary	Min Salary	Max Salary
Executive Office	1	\$250,000	\$250,000	\$250,000
IT/IS	50	\$97,065	\$50,178	\$220,450
Software Engineering	11	\$94,989	\$77,692	\$108,987
Admin Offices	9	\$71,792	\$49,920	\$106,367
Sales	31	\$69,061	\$55,875	\$180,000
Production	209	\$59,954	\$45,046	\$170,500

Key Insight

Production is by far the largest department with 209 employees, but has the lowest average salary at \$59,954. IT/IS pays almost 62% more on average (\$97,065). The salary spread within IT/IS is also huge — from \$50K to \$220K — which suggests very different roles bundled under one department label.

- Worth investigating whether Production's lower pay is contributing to its high attrition — salary and attrition data together tell a stronger story.

3. ATTRITION ANALYSIS BY DEPARTMENT

Business Question: Which departments are losing the most people?

Department	Total Employees	Terminated	Attrition Rate
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Production	209	83	39.7%
Software Engineering	11	4	36.4%
Admin Offices	9	2	22.2%
IT/IS	50	10	20.0%
Sales	31	5	16.1%
Executive Office	1	0	0.0%

Key Insight

Production's 39.7% attrition rate is alarming — nearly 4 in every 10 employees have left. Software Engineering is also high at 36.4%, though the smaller team size makes this harder to act on. Sales at 16.1% is relatively healthy. Overall the company has lost 104 out of 311 employees, which is a 33.4% attrition rate — quite high for any organisation.

- Production and Software Engineering need immediate retention focus — combined they account for the bulk of all terminations.
- Sales' lower attrition despite lower average salary suggests non-monetary factors (culture, manager, flexibility) may matter more.

4. PERFORMANCE SCORE DISTRIBUTION

Business Question: How are employees performing, and does performance relate to salary?

Performance Score	Employee Count	Avg Salary	Avg Satisfaction
Fully Meets	243	\$68,367	3.95
Exceeds	37	\$77,145	4.11
Needs Improvement	18	\$68,408	3.61
PIP (Performance Plan)	13	\$58,971	2.54

Key Insight

The salary gap between top performers and the rest is noticeable but not dramatic — Exceeds earns about 13% more than Fully Meets. What stood out more to me was satisfaction: employees on a Performance Improvement Plan have an average satisfaction of just 2.54 out of 5, compared to 4.11 for top performers. Low satisfaction likely feeds into the attrition numbers we saw earlier.

- Employees on PIP are both the lowest paid and least satisfied — if satisfaction drops low enough, they leave before performance can improve.

5. TOP TERMINATION REASONS

Business Question: Why are employees actually leaving?

Termination Reason	Count
Another position	20
Unhappy	14
More money	11
Career change	9

Hours	8
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Key Insight

The top reason people left was being poached by another position (20 cases), followed by being unhappy (14) and wanting more money (11). Together, unhappiness and money account for 25 out of 104 terminations. These are both addressable — they're not about life circumstances, they're about the employee experience. Career change and hours round out the top 5 and suggest flexibility and growth paths are also factors.

- 25% of terminations come down to unhappiness or pay — both of which are within the company's control to address.

6. GENDER SALARY ANALYSIS

Business Question: Is there a salary gap between male and female employees?

Gender	Headcount	Avg Salary	Min Salary	Max Salary
Female	176	\$67,787	\$45,046	\$250,000
Male	135	\$70,629	\$45,115	\$178,000

Key Insight

There is a salary gap — males earn on average \$2,843 more than females (\$70,629 vs \$67,787). That's about a 4.2% difference. Interestingly, the highest salary in the dataset (\$250,000 — the CEO) is held by a female, which skews the female max upward. Without that outlier, the gap would likely look wider. This deserves a deeper look by role and department to determine if the gap is explained by position differences or is a genuine equity issue.

- A 4.2% average salary gap between genders is worth investigating further — particularly within the same job roles and departments.

7. RECRUITMENT SOURCE EFFECTIVENESS

Business Question: Which recruitment channels bring in the best employees?

Source	Hired	Terminated	Top Performer Rate	Attrition %
Indeed	87	21	14%	24.1%
LinkedIn	76	18	12%	23.7%
Google Search	49	30	4%	61.2%
Employee Referral	31	5	16%	16.1%
Diversity Job Fair	29	16	21%	55.2%
CareerBuilder	23	11	9%	47.8%
Website	13	1	8%	7.7%

Key Insight

Employee Referral is clearly the best channel — only 16.1% attrition and the highest top performer rate among the volume sources at 16%. Google Search looks bad on paper (61.2% attrition) but also only produced a 4% top performer rate, suggesting the quality of candidates from that channel is low.

The company Website has a tiny 7.7% attrition rate which is impressive — those candidates probably sought out the company intentionally and had stronger commitment from day one.

- Referrals should be incentivised — lowest attrition, highest performer rate, and already 31 hires from this channel.
- Google Search and CareerBuilder are producing high attrition at significant volume — worth reconsidering budget allocation to these channels.

8. ABSENTEEISM BY DEPARTMENT

Business Question: Which departments have the highest absence and lateness rates?

Department	Avg Absences	Avg Days Late (30d)	Headcount
Sales	11.55	0.65	31
IT/IS	10.44	0.14	50
Production	10.14	0.47	209
Executive Office	10.00	0.00	1
Software Engineering	8.73	0.36	11
Admin Offices	8.67	0.00	9

Key Insight

Sales has the highest average absences at 11.55 per employee and also the highest lateness rate. Given Sales also has relatively low attrition, this could suggest employees are disengaged — present enough to stay but not fully committed. Production's high absences (10.14) combined with its high attrition paints a picture of a department under stress. Software Engineering and Admin have the healthiest absence numbers.

- Sales' high absences paired with low attrition could indicate disengagement — employees staying but not fully showing up.

9. PERFORMANCE BAND vs SALARY & WELLBEING

Business Question: Do higher performers earn more and feel better about their jobs?

Performance Band	Count	Avg Salary	Avg Satisfaction	Avg Absences
High Performer	37	\$77,145	4.11	10.49
Meets Expectations	243	\$68,367	3.95	10.22
Needs Improvement	31	\$64,450	3.16	10.06

Key Insight

Higher performers are paid more and are more satisfied — that part makes sense. But absences are actually slightly higher among top performers (10.49) than the rest. That was unexpected. It could mean top performers are burning out, or it could just be noise given the small sample size of 37. Either way it's worth monitoring — a burned out high performer eventually becomes an attrition risk.

- The satisfaction gap between High Performers (4.11) and Needs Improvement (3.16) is significant — low performers may need support, not just monitoring.

10. TOP MANAGERS BY TEAM ENGAGEMENT

Business Question: Which managers have the most engaged teams?

Manager	Team Size	Avg Engagement	Avg Satisfaction
Kelley Spirea	22	4.48	3.82
Brandon R. LeBlanc	7	4.35	3.57
Webster Butler	21	4.33	4.05
Simon Roup	17	4.23	3.82
Janet King	19	4.18	3.63

Key Insight

Kelley Spirea leads on engagement (4.48) but Webster Butler's team has the highest satisfaction at 4.05 — suggesting engagement and satisfaction aren't always the same thing. These managers are likely doing something right in terms of how they run their teams. It would be worth having HR document their management practices and use them as a model for lower-performing managers.

- High-engagement managers should be studied — their practices could be used to coach managers whose teams show higher attrition or lower satisfaction.

11. SALARY RANKING WITHIN DEPARTMENTS (WINDOW FUNCTION)

Business Question: How does each employee's salary compare to their department average?

Employee	Department	Salary	Dept Rank	Dept Avg	Diff from Avg
Boutwell, Bonalyn	Admin Offices	\$106,367	#1	\$71,792	+\$34,575
Foster-Baker, Amy	Admin Offices	\$99,351	#2	\$71,792	+\$27,559
Howard, Estelle	Admin Offices	\$49,920	#9	\$71,792	-\$21,872
Zamora, Jennifer	IT/IS	\$220,450	#1	\$97,065	+\$123,385
Foss, Jason	IT/IS	\$178,000	#2	\$97,065	+\$80,935
Voldemort, Lord	IT/IS	\$113,999	#10	\$97,065	+\$16,934

Key Insight

The window function query lets us see salary inequality within each department, not just across them. In IT/IS, the top earner (\$220,450) makes almost 4.4x the department average — a massive spread. In Admin Offices, the range goes from \$49,920 to \$106,367 within the same department. This kind of within-department inequality, if not tied to clear role differences, can cause resentment and attrition.

- Large within-department salary spreads should be reviewed for fairness — especially where employees in similar roles have very different pay.

12. CONCLUSIONS & RECOMMENDATIONS

- 1 Fix Production retention urgently — 39.7% attrition on a 209-person department is unsustainable and expensive.

- 2 Invest in Employee Referral programs — lowest attrition (16.1%), highest performer rate, proven results.
- 3 Review discount and pay practices — 25% of terminations cite unhappiness or money, both are fixable.
- 4 Audit salary equity by gender within the same roles — the 4.2% overall gap needs a closer look.
- 5 Study top managers like Kelley Spirea and Webster Butler — their team practices should inform company-wide management training.